

Esercizio S3 L4

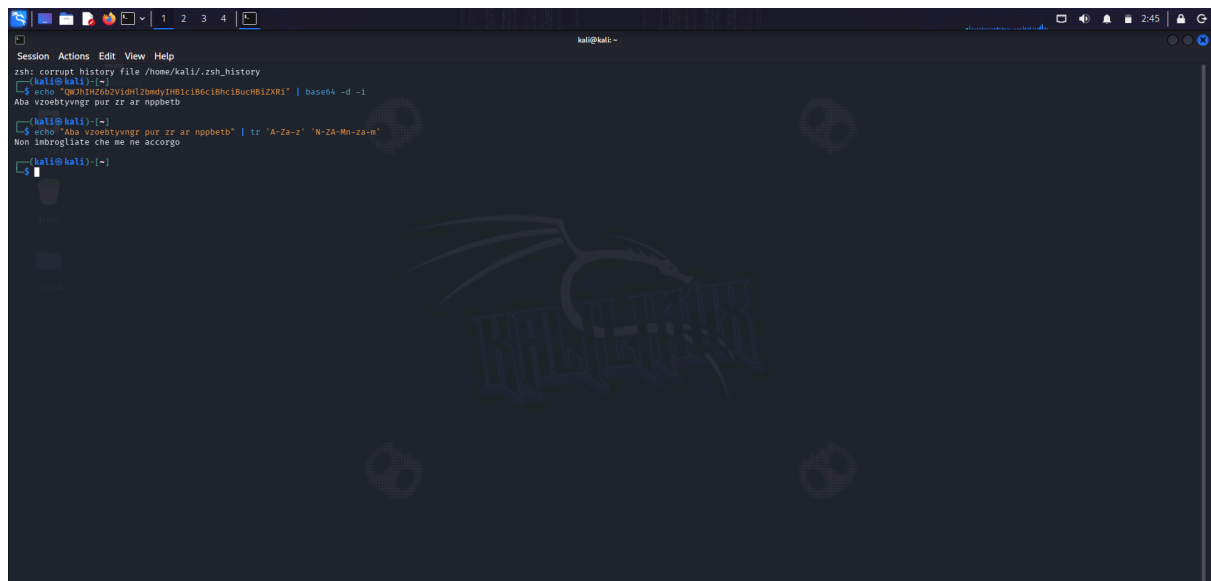
Esercizio 1: "HSNFRGH"

È un **cifrario di Cesare** con scorrimento di 3 lettere indietro

- H = E
- S = P
- N = I
- F = C
- R = O
- G = D
- H = E

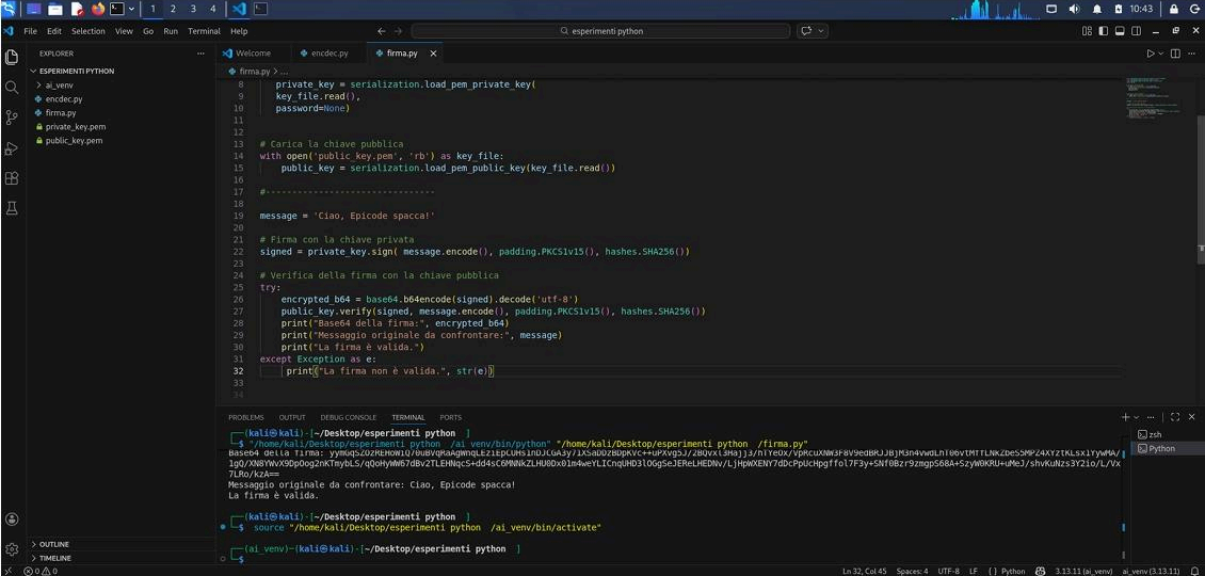
Testo in chiaro: **EPICODE**

Esercizio 2: "QWJhIHZ6b2VidHI2bmdyIHB1ciB6ciBhciBucHBiZX Ri"



```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
kali@kali:~$ echo "QWJhIHZ6b2VidHI2bmdyIHB1ciB6ciBhciBucHBiZX Ri" | base64 -d -i  
Aba vzeobtyvngvr pur zr ar nppbetb  
kali@kali:~$ echo "Aba vzeobtyvngvr pur zr ar nppbetb" | tr 'A-Za-z' 'N-ZA-Mn-za-m'  
Non imbrogliate che me ne accorgo  
kali@kali:~$
```

Esercizio opzionale:



```
File Edit Selection View Go Run Terminal Help
C:\esperimenti\python

EXPLORER
< ESperimentiPython
  > ai_venv
  < encdec.py
  < firma.py
  < private_key.pem
  < public_key.pem

Welcome
encdec.py
firma.py x

8 private_key = serialization.load_pem_private_key(
9     key_file.read(),
10     password=None)
11
12
13 # Carica la chiave pubblica
14 with open('public_key.pem', 'rb') as key_file:
15     public_key = serialization.load_pem_public_key(key_file.read())
16
17 #-----
18 message = 'Ciao, Epcode spacca!'
19
20
21 # Firma con la chiave privata
22 signed = private_key.sign(message.encode(), padding.PKCS1v15(), hashes.SHA256())
23
24 # Verifica della firma con la chiave pubblica
25 try:
26     encrypted_b64 = base64.b64encode(signed).decode('utf-8')
27     public_key.verify(signed, message.encode(), padding.PKCS1v15(), hashes.SHA256())
28     print("Base64 della firma:", encrypted_b64)
29     print("Messaggio originale da confrontare:", message)
30     print("La firma è valida.")
31 except Exception as e:
32     print("La firma non è valida.", str(e))
33
34
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python

[kali@kali]~/Desktop/esperimenti/python
$ /home/kali/Desktop/esperimenti/python/ai_venv/bin/python /home/kali/Desktop/esperimenti/python/firma.py
Base64 della firma: ymeQp0zz6nont10usavvafkgmna5218P0mIndJ5A5y/11C80260pvc+HwPvgpJ/28Dx13m3j3/011Y0X/9pGCUWw3F8V9edBkJ8JMSn4wvdlN1WvVMTTLWkZde53MP24XZTLKs1YymWw/1gQ/X08YwVw50p0og2nKtybLS/qQohYm67Dbv2TLEHlqC5+dd4sC0WwKZLHURDx01ndweYLIcnqHD310GgSe3EReLHEDWw/LJHqX0ENY7d0cPpUcHpgfFo17F3y+SNf0Bz79zngp568A+5zyW0KRU+MeJ/shvKulZs3Y21o/L/Vx
7L0u/Kz==
Messaggio originale da confrontare: Ciao, Epcode spacca!
La firma è valida.

[kali@kali]~/Desktop/esperimenti/python
$ source "/home/kali/Desktop/esperimenti/python/ai_venv/bin/activate"

(ai_venv) [kali@kali]~/Desktop/esperimenti/python
$
```